By employing hybrid CSA and HCA, along with precision optimizations on computation and storage, DeepSeek-V4 series achieve significantly lower inference FLOPs and a substantially reduced KV cache size compared with DeepSeek-V3.2, especially in long-context settings. The right part of Figure 1 demonstrates the estimated single-token inference FLOPs and accumulated KV cache size of DeepSeek-V3.2 and DeepSeek-V4 series. In the scenario of 1M-token context, even DeepSeek-V4-Pro, which has a larger number of activated parameters, attains only 27% of the single-token FLOPs (measured in equivalent FP8 FLOPs) and 10% of the KV cache size relative to DeepSeek-V3.2. Furthermore, DeepSeek-V4-Flash, with its smaller number of activated parameters, pushes efficiency even further: in the 1M-token context setting, it achieves only 10% of the single-token FLOPs and 7% of the KV cache size compared with DeepSeek-V3.2. Additionally, for DeepSeek-V4 series, the routed expert parameters utilize FP4 precision. While the peak FLOPs for FP4 $\times$ FP8 operations are currently the same as FP8 $\times$ FP8 on existing hardware, they can theoretically be implemented to be 1/3 more efficient on future hardware, which will further enhance the efficiency of DeepSeek-V4 series.

During pre-training, we train DeepSeek-V4-Flash on 32T tokens and DeepSeek-V4-Pro on 33T tokens, respectively. After pre-training, these two models can natively and efficiently support 1M-length contexts. In our internal evaluations, DeepSeek-V4-Flash-Base already surpasses DeepSeek-V3.2-Base across a majority of benchmarks with its more parameter-efficient design. DeepSeek-V4-Pro-Base further extends this advantage to set a new performance standard among DeepSeek foundation models, achieving comprehensive superiority across reasoning, coding, long-context, and world knowledge tasks.

The post-training pipeline of DeepSeek-V4 series features a two-stage paradigm: the independent cultivation of domain-specific experts, followed by unified model consolidation via on-policy distillation (Gu et al., 2024; Lu and Lab, 2025). Initially, for each target domain — such as mathematics, coding, agent, and instruction following — a separate expert model is trained independently. The base model first undergoes Supervised Fine-Tuning (SFT) on high-quality, domain-specific data to establish foundational capabilities. Subsequently, Reinforcement Learning (RL) is applied using Group Relative Policy Optimization (GRPO) (DeepSeek-AI, 2025), which further optimizes the model for domain-aligned behaviors guided by reward models tailored to specific success criteria. This phase yields a diverse set of specialized experts, each excelling in its respective field. Finally, to integrate these distinct proficiencies, a single unified model is trained through on-policy distillation, wherein the unified model acts as the student learning to optimize the reverse KL loss with teacher models.

Summary of Core Evaluation Results

- **Knowledge**: In assessments of broad world knowledge, DeepSeek-V4-Pro-Max, the maximum reasoning effort mode of DeepSeek-V4-Pro, significantly outperforms leading open-source models on the SimpleQA (OpenAI, 2024d) and Chinese-SimpleQA (He et al., 2024) benchmarks. Regarding educational knowledge — evaluated via MMLU-Pro (Wang et al., 2024b), HLE (Phan et al., 2025), and GPQA (Rein et al., 2023) — DeepSeek-V4-Pro-Max shows a marginal lead over its open-source counterparts. DeepSeek-V4-Pro-Max has significantly closed the gap with the leading proprietary model, Gemini-3.1-Pro, despite still trailing it in these knowledge-based evaluations.
- **Reasoning**: Through the expansion of reasoning tokens, DeepSeek-V4-Pro-Max demonstrates superior performance relative to GPT-5.2 and Gemini-3.0-Pro on standard reasoning benchmarks. Nevertheless, its performance falls marginally short of GPT-5.4 and Gemini-3.1-Pro, suggesting a developmental trajectory that trails state-of-the-art frontier models by approximately 3 to 6 months. Furthermore, DeepSeek-V4-Flash-Max achieves comparable